STAFF REVIEW · H446 015 source models

Separate from the eight-page pupil workbook. This copy is in a public repository and is NOT access controlled. Check technical correctness before use.

Printed question mapping: Booklet Q1 = original online Q1; Booklet Q2 = original online Q4

Do Now — source answers

1. DrawBoard().

2. Row 1, column 2.

3. Traverse every row and column.

4. [1,1].

I Do — source worked steps

1. Subtract 1 — CODE:
move - 1 = 7 — Board indexes begin at 0.

2. Integer divide by 3 — CODE:
7 / 3 = 2
row = 2 — Integer division discards the remainder.

3. Use MOD 3 — CODE:
7 % 3 = 1
col = 1 — The remainder identifies the column.

4. Place the player — CODE:
board[2,1] = player; — Move 8 maps to row 2, column 1.

We Do — guided CHECK

1. Complete the same calculation
CODE:
row = (6 - 1) / 3 = 1
col = (6 - 1) % 3 = 2

2. Fill the same condition
CODE:
if (move < ? || move > ?)
{
return false;
}
CODE:
if (move < 1 || move > 9)
{
return false;
}

3. Fill the same condition
CODE:

if (board[row,col] == ? ||
board[row,col] == ?)
{
return false;
}
CODE:
if (board[row,col] == "X" ||
board[row,col] == "O")
{
return false;
}

4. Fill the same code
CODE:
if (player == "X")
return ?;
else
return ?;
CODE:
if (player == "X")
return "O";
else
return "X";

You Do — source mark schemes

1. Map a move
CODE:
int row = (move - 1) / 3;
int col = (move - 1) % 3;
Check: 1→[0,0], 5→[1,1], 9→[2,2].

2. Write MakeMove()
CODE:
public static bool MakeMove(int move, string player)
{
if (move < 1 || move > 9)
return false;

int row = (move - 1) / 3;
int col = (move - 1) % 3;

if (board[row,col] == "X" || board[row,col] == "O")
return false;

board[row,col] = player;
return true;
}
Check: Invalid moves leave the board unchanged.

3. Write SwitchPlayer()
CODE:

```
public static string SwitchPlayer(string player)
{
if (player == "X")
return "O";
else
return "X";
}
```

Check: The method alternates correctly.

4. Make alternating turns work
CODE:

```
string currentPlayer = "X";
int movesMade = 0;

while (movesMade < 9)
{
DrawBoard();
Console.Write("Player " + currentPlayer + ", choose 1-9: ");
int move = int.Parse(Console.ReadLine());

if (MakeMove(move, currentPlayer))
{
movesMade++;
currentPlayer = SwitchPlayer(currentPlayer);
}
else
{
Console.WriteLine("Invalid move - try again.");
}
}

DrawBoard();
```

Check: Checkpoint reached: X and O can take valid alternating turns. Win/draw detection comes next lesson.

Exit Ticket — source answer

1. False. An occupied square is invalid. Ask the same player to choose another square.